



Khalifa Bin Zayed School
Physics Department - Grade 10 Advance
Module 17, Lesson 2: Diffraction Analysis Lab

Subject: Physics
Grade: 10 ADV

Student Identification

Name:

Class: 10 ADV 1

Date: 11/5/2026

Objective

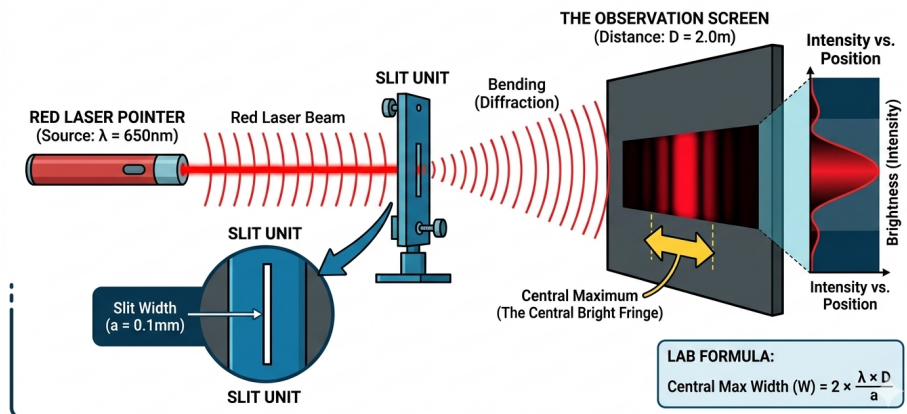
To verify the single-slit diffraction formula $W = \frac{2\lambda D}{a}$ by measuring the central maximum width using a virtual precision ruler.

Theory & Setup

W = Total Width of Central Maximum | λ = Wavelength

a = Slit Width | D = Screen Distance

GRADE 10 LAB: SINGLE SLIT DIFFRACTION



Observation Table

Tr.	λ (nm)	a (mm)	D (m)	Measured W (mm)	Theor. W (mm)	% Error
1	452	0.20	1.5	5	6.8	1.8
2	452	0.20	2.5	10	11.3	1.3
3	452	0.20	5	21	22.6	1.6

Average % Error: 1.6

Analysis Questions

1. Concept: If you switch from a Red Laser to a Blue Laser, does the central fringe get **wider** or **narrower**? (Justify using the formula).

It Gets narrower because W is directly proportional to λ . so if we move from a greater to a smaller wavelength the central fringe will get narrower

2. Experiment: Why is it more accurate to measure the total width (W) rather than just the distance from the center to the first dark fringe?

It Gets narrower because W is directly proportional to λ . so if we move from a greater to a smaller wavelength the central fringe will get narrower

It helps prevent human error

Conclusion

It Gets narrower because W is directly proportional to λ . so if we move from a greater to a smaller wavelength the central fringe will get narrower

more accurate because humans can be slightly off when measuring which can cause the answer to be Slightly different and less accurate |